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Mapping neurodevelopmental diversity in executive function

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ABSTRACT

Executive function, an umbrella term used to describe the goal-directed regulation of thoughts, actions, and emotions, is an important dimension implicated in neurodiversity and established malleable predictor of multiple adult outcomes. Neurodevelopmental differences have been linked to both executive function strengths and weaknesses, but evidence for associations between specific profiles of executive function and specific neurodevelopmental conditions is mixed. In this exploratory study, we adopt an unsupervised machine learning approach (self-organising maps), combined with k-means clustering to identify data-driven profiles of executive function in a transdiagnostic sample of 566 neurodivergent children aged 8–18 years old. We include measures designed to capture two distinct aspects of executive function: performance-based tasks designed to tap the state-like efficiency of cognitive skills under optimal conditions, and behaviour ratings suited to capturing the trait-like application of cognitive control in everyday contexts. Three profiles of executive function were identified: one had consistent difficulties across both types of assessments, while the other two had inconsistent profiles of predominantly rating- or predominantly task-based difficulties. Girls and children without a formal diagnosis were more likely to have an inconsistent profile of primarily task-based difficulties. Children with these different profiles had differences in academic achievement and mental health outcomes and could further be differentiated from a comparison group of children on both shared and profile-unique patterns of neural white matter organisation. Importantly, children's executive function profiles were not directly related to diagnostic categories or to dimensions of neurodiversity associated with specific diagnoses (e.g., hyperactivity, inattention, social communication). These findings support the idea that the two types of executive function assessments provide non-redundant information related to children's neurodevelopmental differences and that they should not be used interchangeably. The findings advance our understanding of executive function profiles

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and their relationship to behavioural outcomes and neural variation in neurodivergent populations.

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1. Introduction

The transdiagnostic revolution in neurodevelopmental conditions emphasises a move away from diagnostic systems and towards identifying broad dimensions that characterise neurodiversity (Astle et al., 2022). One such dimension, which captures characteristics associated with multiple neurodevelopmental differences (Bloemen et al., 2018), can be a barrier to learning (Holmes et al., 2021), is a well-established and malleable predictor of multiple adult outcomes (Diamond, 2013), and is linked to neural differences in white matter organisation (Bathelt et al., 2018; Baum et al., 2017) is executive function. The term executive function is used widely to describe the processes supporting the volitional control of cognition and behaviour, and in this way, it is sometimes referred to as a system of cognitive control. In this exploratory study, we adopt a transdiagnostic data-driven approach to delineate profiles of executive function among a large neurodevelopmentally neurodivergent sample using measures designed to capture two conceptually distinct aspects of executive function: performance-based tasks designed to tap higher-order cognitive skills (Miyake et al., 2000), and behaviour ratings suited to capturing the application of cognitive control in everyday contexts (Doebel, 2020). We then explore whether and how these different profiles (i.e., groups of children with different profiles identified in a data-driven manner), are associated with differences in neurodevelopmental diagnostic categories and dimensions, academic and mental health outcomes, and neurobiology.

Executive function can be conceptualised in two ways that map broadly onto the ways in which they are measured (Malanchini et al., 2018; Toplak et al., 2013). The first views executive function as a set of higher-order cognitive skills that enable flexible thinking (Miyake et al., 2000). There are many different accounts of executive function that vary in whether they consider it to be a unitary construct or a set of related skills or components (for a review, see Baggetta & Alexander, 2016). There are also differences in the sets of skills included under the ‘executive function umbrella’, but these are broadly considered to include shifting/switching, inhibition, working memory, planning, and (sustained) attention (Baggetta & Alexander, 2016; Carlson et al., 2004; Deodhar & Bertenthal, 2023; Friedman & Miyake, 2017; Garon et al., 2008; Miyake et al., 2000; St Clair-Thompson & Gathercole, 2006). These are typically all indexed by accuracy or response times on performance-based tasks, such as memory span tasks that index working memory performance (e.g., Automated Working Memory Assessment, Alloway, 2007), tasks that require goal-oriented planning (e.g., Tower of London, Shallice, 1982), semantic incongruence or cognitive flexibility tasks that

assess shifting/switching (e.g., Number-Letter sequencing, Delis et al., 2001; Stroop, 1935), and tasks that require focused attention in the face of distraction over lengthy periods that measure sustained attention and response inhibition (e.g., Continuous Performance Test, Conners, 1992). There are many other task-based measures of executive function, and multiple variants of the examples provided here (see Strauss et al., 2006 for a comprehensive list of task-based measures of executive function), but what they have in common is that they are administered individually under controlled experimental conditions requiring effortful cognitive control. In this way, therefore they can be viewed to reflect optimal (state-like) rather than typical (trait-like) performance.

The second view of executive function integrates knowledge, beliefs, and values alongside cognitive, motor, and perceptual control, and makes explicit that these factors come together in the service of goals in specific settings (Doebel, 2020). It refers to the individual use of executive skills in particular situations, rather than a general “capacity process” such as inhibition that is applied across all situations. To illustrate the distinction, Doebel (2020) uses the example of a child inhibiting the impulse to hit another child who has taken their toy. In this situation, the child brings to bear knowledge and beliefs about hitting another child (e.g., socially acceptable alternatives to hitting, beliefs about being scolded for engaging in hitting behaviour) when inhibiting a response or action. In this way, executive function is viewed as inseparable from the task goals, context, and mental content that come into play in specific situations. Understanding executive function as tied to the activities a person is completing in their everyday activities mirrors the second way in which executive function is measured – using rating scales to index the ability to coordinate multiple processes in everyday problem-solving situations that draw on knowledge, beliefs, and values. Examples of such rating scales include the Behaviour Rating Index of Executive Function (BRIEF; Gioia et al., 2000) and the Working Memory Rating Scale (Alloway et al., 2009). These, and similar questionnaires, require self or observer ratings of the frequency of everyday executive function behaviours in different contexts. They were originally developed to provide an ecologically valid measure of executive function that captures real-world functioning. Questionnaire responses typically reflect behavioural observations over extended periods of time, and may therefore reflect more trait-like or stable processes and provide a more accurate assessment of typical performance (Doebel & Müller, 2023; Holochwost et al., 2023; Malanchini et al., 2018). This may explain why rating-based assessments are often observed to be more psychometrically reliable than cognitive tasks (Hedge et al., 2018; Pezzoli et al., 2023).

Two lines of evidence suggest that measures of executive function used in these two domains – cognitive processes and

skills involved in goal-oriented behaviour in everyday situations – capture different information and are dissociable. First, there is little convergent validity between the measurement types tapping each domain (Dang et al., 2020). If performance-based and rating-based measures of executive function were fully overlapping and capturing the same information, they should be highly correlated. Yet concurrent and predictive validity studies consistently indicate that they are, at best, weakly correlated (e.g., Gerst et al., 2017; Nin et al., 2022; Soto et al., 2020; Tamm & Peugh, 2019; Toplak et al., 2013), and that these relationships remain small when using latent variable approaches (Snyder et al., 2021) and/or when controlling for mono-method bias (e.g., that performance-based measures predict other performance-based measures, and rating-based measures predict other rating-based measures Soto et al., 2020). In a review, Toplak et al. (2013) concluded that the two types of measurement capture different information, with rating-based measures capturing goal-oriented successes in typical everyday settings and performance-based measures capturing processing efficiency under optimal conditions.

Second, they make independent contributions to clinical, academic, and life outcomes. Difficulties with executive function are associated with multiple neurodevelopmental conditions, including attention deficit hyperactivity disorder (ADHD), specific learning difficulties, and autism (e.g., Benallie et al., 2021; Holmes et al., 2014; Loe & Feldman, 2007; McClain et al., 2022; Willcutt et al., 2005), they can be a barrier to learning (Holmes et al., 2021; Peng & Fuchs, 2016; Soto et al., 2021; Yeniad et al., 2013), and are also linked to poor mental health outcomes (e.g., Bloemen et al., 2018). Substantial evidence suggests that the different measurement methods of executive function make unique and separable contributions to these outcomes (Gerst et al., 2017; Soto et al., 2020). In terms of academic achievement, both rating- and performance-based measures predict outcomes, but the relationships are typically stronger and more consistent for performance-based measures (Gerst et al., 2017; Malanchini et al., 2018; Soto et al., 2020). Similarly, for mental health outcomes, rating- and task-based measures of executive function explain independent variance in both internalising and externalising difficulties (Eisenberg et al., 2019; Ellingson et al., 2019; Friedman et al., 2020; Friedman & Gustavson, 2022), with stronger relationships typically reported between rating-based measures and mental health (Friedman et al., 2020), although mono-method bias is rarely controlled for in such comparisons.

The correspondence between ratings of executive function behaviours and performance on cognitive tests is also low in neurodivergent groups. For example, in both children and adults with ADHD, executive function profiles captured by performance-based and rating-based measures have little overlap (Biederman et al., 2008; Krieger & Amador-Campos, 2018; Tan et al., 2018). Moreover, both types of assessments make distinct contributions when predicting occupational outcomes in people with ADHD (Barkley & Murphy, 2010, 2011). Similar differences between the two types of assessment have been reported for other neurodevelopmental conditions (Gómez-Pérez et al., 2016; Gross et al., 2015). In a transdiagnostic study, Williams et al. (2022) found that a substantial proportion of a large sample of neurodivergent

children had what they termed as an inconsistent executive function profile, defined as difficulties reported on rating-based assessments of executive function that were not evident on performance-based tasks.

1.1. Current study

The aim of the current study was to adopt a data-driven transdiagnostic approach to delineate profiles of executive function among a large neurodevelopmentally neurodivergent sample. Previous studies exploring executive function in neurodivergent children have typically either contrasted executive function performance between groups (e.g., those with a diagnosis compared to those with a different, or no diagnosis) across a range of performance-based measures (e.g., Geurts et al., 2004; Holmes et al., 2014), explored how well executive function rating scales and performance tests predict outcomes separately (e.g., Gerst et al., 2017; Soto et al., 2020), or tested relationships between different types of executive function measurement (e.g., Nin et al., 2022; Tamm & Peugh, 2019). Studies applying strict diagnostic thresholds or using case–control designs fail to capture broad populations of neurodivergent children and tend to overemphasise the differences between diagnostic groupings (Astle et al., 2022). For these reasons, here, we used a data-driven approach and focused on a transdiagnostic sample that included a range of children who had been identified as having additional needs by health and education practitioners, irrespective of their diagnostic status. Classifying individuals into homogenous data-driven groupings, based on executive functioning rather than diagnostic status, provides a way to more accurately map phenotypes to outcomes and neurophysiology (Bathelt et al., 2018; Vaidya et al., 2020). To do this, we used a self-organising map algorithm to map the multidimensional space of a broad range of performance- and rating-based assessments of executive function, representing how children group together based on their executive function profiles. We then used data-driven clustering to delineate subgroups of children presenting with different executive function profiles and explored how these related to the two dissociable aspects of executive function. We also investigated whether the groups differed in terms of neurodevelopmental diagnostic categories (ADHD, autism, etc.) and dimensions that capture neurodiversity (hyperactivity, inattention, communication). Finally, we explored how these groups related to learning, mental health, and neural white matter organisation. We focussed on white matter organisation because there is substantial literature supporting its links to both executive function (Bathelt et al., 2018; Baum et al., 2017) and neurodevelopmental conditions and dimensions (Ameis et al., 2016; Beare et al., 2017).

The study was fully exploratory, and we did not formulate a hypothesis about the number of groups or the phenotypic and neural features that would differentiate them. Instead, we used a data-driven approach to address the following questions: 1) can we identify subgroups of children with different profiles of executive function; 2) if so, do these groups differ in terms of the two dissociable aspects of executive function; and 3) how do these groups relate to neurodevelopmental diagnostic categories/characteristics, academic and mental

health functioning, and neural white matter organisation? No part of the study procedures or analysis plans was preregistered prior to this research being conducted.

2. Method

2.1. Participants

Participants were drawn from the Centre for Attention, Learning, and Memory (CALM) cohort. Recruitment details and testing procedures are described in the study protocol (Holmes et al., 2019). Here, we provide a short overview and report on how we determined the final sample size for this analysis, all data exclusions, whether inclusion/exclusion criteria were established before data analysis, all manipulations, and all measures used in this analysis. Broadly, children aged 5–18 years were referred to the study by health and educational professionals for difficulties with attention, learning, and/or memory regardless of diagnostic status. Most referrals came from education services, which included special educational needs coordinators, specialist teachers, educational psychologists, and head teachers (63 %). A smaller proportion were referred from health services, which included clinical psychologists, psychiatrists, and paediatricians (33 %). A minority of the sample were referred by speech and language therapists (4 %). Participants aged 8 years and above were included in the current study. Younger children were excluded before data analyses because some of the executive function tasks were not standardised for children under 8 years old and were therefore not administered to them. The sample for this study included 566 children (45 % with at least one diagnosis) with an average age of $M = 10.55$ years, $SD = 2.02$ ($N_{boys} = 380$, $N_{girls} = 186$). The most common diagnoses were ADHD ($N = 156$, of whom $N_{medicated} = 88$), autism ($N = 46$), and dyslexia ($N = 47$). Smaller numbers of children had dyscalculia, developmental language disorder, dyspraxia, global developmental delay, sensory processing disorder, Tourette's syndrome, and mental health conditions (see Holmes et al., 2021 for a full breakdown of diagnoses in this cohort). Demographic information is not reported here by diagnostic category for several reasons. First, diagnostic comparisons have been reported previously for some of the larger diagnostic groups in this cohort (Holmes et al., 2021). Second, diagnostic groupings were not discrete – many children with diagnoses had more than a single diagnosis and these high comorbidity rates meant that it was not feasible to make meaningful comparisons due to small group sizes. We instead report a comparison between children who had any diagnosis and those who did not across the executive function measures in the [Supplementary Materials \(Table S2\)](#). This revealed that those with a diagnosis tended to be older and experienced more difficulties on rating-based assessments of executive function relative to non-diagnosed participants. No differences were observed on any of the performance-based measures.

Ethnicity information was not collected as part of the original CALM study, but it was requested retrospectively as part of an enrichment phase of the project to which only a subsample of the original cohort responded ($N = 189$, of which 176 identified as 'White'; 12 as 'Mixed/Multiple Ethnic Groups';

and 1 as Black/African/Caribbean/Black British). As a proxy for socioeconomic status, the index of multiple deprivation (a measure that ranks relative levels of deprivation across different areas of England) based on participant's postcodes was used. These data were available for 517 participants ($M = 20,455$, $SD = 8,222$, $min = 155$, $max = 32,803$). The national ranking ranges from 1 (most deprived) to 328,444 (least deprived), suggesting that our sample included participants living in varying levels of deprivation. Ethical approval was obtained by the National Health Service (REC: 13/EE/0157). Parents/caregivers gave written consent and children gave verbal assent to participate.

2.2. Measures

Children completed a range of assessments of learning and cognition and a questionnaire asking about their mental health. Parents/caregivers completed questionnaires about the child's behaviour and communication skills. All assessments were administered by a trained researcher at the CALM clinic, following standardised administration procedures documented in the test manuals. Task details and administration procedures are available in the study protocol, and psychometric properties have been previously reported in other publications using this cohort (Holmes et al., 2021; Mareva & Holmes, 2019). A summary of each assessment included in this study is provided below. For all assessments, norms for average age-typical levels of functioning were available and therefore age-referenced scores were used in all analyses. Legal copyright restrictions prevent public archiving of most assessments listed below (except the Revised Child Anxiety and Depression Scale), which can be obtained from the copyright holders in the cited references.

2.2.1. Executive function

Task-based measures – Tasks tapping executive functioning were chosen from the broader neuropsychological battery used in the CALM study (Holmes et al., 2019). These included four tasks from the Automated Working Memory Assessment (AWMA; Alloway, 2007), tapping verbal and visuospatial short-term/working memory: forward digit recall, backward digit recall, dot matrix, and Mr X. Two tasks from the Delis-Kaplan Executive Function System test battery (D-KEFS, Delis et al., 2001) were included: the Tower task, commonly used to assess planning skills, and the Trail-making number-letter sequencing task, a Stroop-like task indexing shifting. Two tasks from the Test of Everyday Attention for Children 2 (TEA-Ch2, Manly et al., 2016) were also included: the switching task, Reds, Blues, Bags and Shoes (RBBS), and the Vigil task tapping sustained attention.

Rating-based measures – Parents/caregivers completed the BRIEF (Gioia et al., 2000), an 80-item rating scale covering eight domains: Inhibition, Shifting, Emotional control, Initiation, Working memory, Planning, Organisation, and Monitoring.

Age-standardised scores derived from performance and rating-based measures were a combination of T-, scaled and standardised scores and were converted to age-referenced z-scores to be on the same scale (a score of zero corresponded to age-typical performance, a score of one represented 1 SD

above the age-expected mean, and -1 represented 1 SD below the age-expected mean).

2.2.2. Learning

The Word Reading and Numerical Operations subtests of the Wechsler Individual Achievement Test II (WIAT-II; Wechsler, 2005) were used to assess children's reading and mathematical skills.

2.2.3. Mental health

The total anxiety and depression score from the Short Youth-Report Revised Child Anxiety and Depression Scale <https://dev-rcads-ucla-site.pantheon.io/sites/default/files/2023-06/RCADS25-Youth-English-2018.pdf> (RCADS, Chorpita et al., 2005) was used to index child-reported internalising difficulties. This questionnaire was introduced into the CALM protocol mid-way through the study, meaning data were only available for 269 participants.

2.2.4. Diagnostic categories and dimensions of neurodiversity

Neurodevelopmental diagnoses were reported by the referring professional and confirmed by parents/caregivers at the time of referral to the CALM study. Ratings on dimensions of neurodiversity that are core features of neurodevelopmental conditions that were included in the CALM protocol were included in the current study. These were ratings provided by the child's parent/caregiver using the following scales: inattention and hyperactivity/impulsivity (dimensions of ADHD measured by the Conners-3 Parent Short Form Rating Scale (Conners, 2008); communication skills (dimension of developmental language disorder measured by the Child Communication Checklist-2, CCC-2, Bishop, 2003), and social communication and interests (core dimensions of autism, measured by the CCC-2, Bishop, 2003).

2.3. Neuroimaging

Neuroimaging data was available for a subset of the cohort. The current analysis included 248 referred children ($M_{age\ scan} = 10.92$, $SD = 2.07$, 70 % male) and 77 children ($M_{age\ scan} = 10.75$, $SD = 2.01$) who formed a comparison group. The children in the comparison group were recruited from the same schools attended by those who were referred, but themselves had not been referred. Children forming this group were not screened for psychiatric/neurodevelopmental conditions, consistent with the transdiagnostic approach. They were included as a socio-demographically matched group of non-referred children who attended the same schools as those who were referred. This non-referred group was included in the neuroimaging analyses for the purposes of data interpretation – unlike the cognitive and behavioural assessments, neuroimaging metrics do not allow for a straightforward interpretation of age-typical levels of functioning. The acquisition and pre-processing steps of the diffusion scans have been reported elsewhere (Mareva et al., 2023) and are briefly described in the [Supplementary Materials](#). In the final step, whole-brain white matter connectivity matrices (i.e. connectomes) were constructed for each child based on the Brainnetome atlas (Fan et al., 2016). The number of streamlines intersecting each pairwise

combination of regions ($N = 246$) was estimated and transformed into a 246×246 streamline matrix.

2.4. Analysis plan

Analyses were conducted in four steps: 1) missing data estimation; 2) self-organising map (SOM) and data-driven clustering; 3) comparison of profiles across executive function, neurodevelopmental, academic, and mental health functioning based on a series of chi-squares and t-tests; 4) pairwise comparisons and partial least squares discriminant analysis (PLSDA) to explore patterns of neural white matter organisation that distinguish the data-driven groups from the non-referred participants who were included in the neuroimaging cohort. Details for each step of the analysis are outlined below.

2.5. Missing data

Missing data were estimated only for the assessments used in the SOMs (see [Table S1](#) for descriptive statistics before imputation). For all other assessments used in the validation analyses, missing data was not estimated to ensure it was independent and suitable for external validation. Data was complete for 59 % of participants. Boys and girls were equally likely to have missing data ($\chi^2 = .22$, $p = .64$) and so were participants with and without a diagnosis ($\chi^2 = .03$, $p = .87$). Missing data was estimated via the random forest nonparametric imputation procedure implemented in R package *missForest* (Stekhoven & Bühlmann, 2012). Participant sex and age were also added to the imputation model to improve estimates.

2.6. Self-organising maps (SOMs)

Self-organising maps (SOMs) are a specific type of artificial neural network designed for noise reduction and the analysis of complex, high-dimensional data in a simplified, lower-dimensional space. Unlike other data-reduction methods such as factor analysis, SOMs do not aim to determine the optimal number of underlying dimensions in the data. Instead, they reduce data complexity by mapping it onto a topological space of nodes, in our case organised along two spatial dimensions. In this space, individuals are assigned to nodes based on similarities across input variables (i.e., 16 executive function measures) expressed as nodal weights. In simpler terms, SOMs group together individuals who share similar input data on this topological map. This means that on the final map, children with similar profiles will be located on nodes closer to each other in space. Subsequently, each assessment from the input data can be projected onto this map, allowing us to gain insights into how specific features differentiate children in various areas of the map. A conceptual overview of the implementation used here is provided in the [Supplementary Materials](#) and more details are available in [Wehrens and Buydens \(2007\)](#).

2.7. Data-driven clustering

The SOM algorithm is a valuable tool for addressing challenges associated with analysing multidimensional data, a

problem known as the ‘curse of dimensionality’ (Dalmajer et al., 2022). It accomplishes this by representing data in a continuous, two-dimensional grid of nodes, where proximity in space reflects similarity. Assuming that our study population is diverse, dividing this map into sections should reveal groups of individuals with similar profiles, distinct from those located elsewhere on the map. The SOM itself doesn’t explicitly define these sections by drawing boundaries between nodes; an additional analysis step is required for this purpose. There isn’t a fixed rule for determining how many sections the map should be divided into. To find the optimal number of groups, we employed an additional data-driven approach, in which we used the nodal weight values obtained from the SOM and submitted them to k-means clustering. This method was chosen for its non-parametric nature and because it makes fewer assumptions about the data than other clustering methods, meaning it aligned well with the exploratory nature of the analysis. Following good practice recommendations, before clustering, the nodal weights were reduced using uniform manifold approximation and projection (Bathelt et al., 2021; Dalmajer et al., 2022). The optimal number of clusters was chosen based on a consensus approach implemented in the R-package NbClust (Charrad et al., 2014). This method uses 30 indices for determining the number of clusters and converges on the best clustering scheme across the different results obtained by combinations of the number of clusters, distance measures, and clustering methods (full description of these metrics is beyond the scope of this manuscript but for a full list and detailed description please see Charrad et al., 2014). This provided us with clusters of children based on the nodes they were assigned to in the original mapping. To check the clustering, each cluster distribution was plotted on the original map with the expectation that all cluster members ought to sit on neighbouring nodes within the original map.

2.8. Cluster characterisation & comparisons

To characterise the data-driven groups we compared their average scores on each of the assessments used in the SOM in a series of t-tests. False discovery correction was used to handle multiple comparisons at each level of analysis (e.g., data used in SOM, behavioural data not included in SOM, neural data). We also validated the data-driven clusters by testing whether group differences generalised to data not included in the SOM – in this case, learning assessments and self-reported mental health difficulties, neurodevelopmental diagnostic categories, and parent-reported dimensions of neurodiversity. For continuous data, t-tests were conducted using false discovery rate as a correction for multiple comparisons, and for categorical data, chi-square tests were used (e.g., to compare the number of children with a particular diagnosis in each subgroup relative to the overall distribution across the whole sample).

2.9. Neuroimaging comparisons

The neuroimaging analyses explored three levels of white matter connectome organisation: global, module, and regional hubs. For the first two, we used a series of non-

parametric comparisons contrasting the non-referred comparison group to the data-driven groups using false-discovery rate correction. This approach was favoured as it allowed us to explore both shared and subgroup-unique differences relative to the comparison group with a minimum number of comparisons. To explore differences in regional hub organisation we used a PLSDA (Brereton & Lloyd, 2014), aiming to derive components of regional hubs that best explain group membership across the data-driven profiles and the non-referred comparison sample. Prior to analyses, the effects of age and in-scanner motion (average frame displacement) were regressed from each metric using a robust estimation approach.

At the global level, we focused on connectome efficiency and the average participation coefficient. Global efficiency describes the potential for information exchange in the connectome and is the average inverse distance from any one brain region to another (Sporns et al., 2007). The participation coefficient measures the proportion of possible connections each region has with regions from other modules. It is considered an index of modular segregation, with lower values indicating more segregation: regions with a high participation coefficient have strong connections to many modules, while regions with a low participation coefficient have strong connections to fewer modules (Baum et al., 2017). A global participation coefficient was derived by averaging the participation coefficient estimated for each one of the 246 regions included in the Brainnetome atlas. These metrics were chosen because they have been previously linked to performance on tests of executive function in developing populations (Baum et al., 2017; Koenis et al., 2015).

To derive module organisation, we grouped the 246 regions of the Brainnetome atlas a priori into their corresponding functional networks as defined by Yeo et al. (2011). Subcortical nodes were also grouped together. This modular parcellation is based on independent functional data but has been previously shown to be a good representation of the organisation seen in the white matter connectome in developing populations (Baum et al., 2017). We then estimated modular strength for each one of the 7 networks and the subcortex. Significantly different results were followed-up to explore whether differences were driven by changes in within-module connectivity, between-module connectivity, or both.

Finally, PLSDA was chosen as a tool to explore links between regional connector hub strength and group membership. PLSDA is a statistical technique used to find relationships between two sets of variables: a predictor set and a response set. In our case these were connector hub strength and group membership respectively. The goal of PLSDA was therefore to identify the linear combination of connector hubs that best explains the variation in group membership. We focused on connector hubs because they play an important role in network specialisation across development and support the development of executive function (Baum et al., 2017; Jones et al., 2021; Zink et al., 2021). In accordance with prior work, connector hubs were defined as regions which were above the 70th percentile on both betweenness centrality and participation coefficient (Jones et al., 2021). The analysis was implemented in the R package mixOmics (Rohart et al., 2017). The model was evaluated by

using 5-fold cross-validation repeated 50 times and the number of components to retain was chosen based on the balanced error rate metric, accounting for the uneven number of participants across groups ($N_{\text{cluster } 1} = 95$, $N_{\text{cluster } 2} = 74$, $N_{\text{cluster } 3} = 79$). The contribution of each hub (i.e., component loadings) to the PLSDA components was then evaluated using a bootstrap procedure ($N = 1000$), where a Procrustes rotation was applied to align the factors across iterations (Krishnan et al., 2011). This was done to test whether the bootstrapped confidence interval passed zero, and thus to establish which hubs reliably load on the PLSDA components. Group differences in component scores were compared using a non-parametric permutation procedure testing whether they significantly differed from chance when group labels were permuted 1000 times.

3. Results

3.1. SOMs and clustering

Performance across all measures used in the SOM is presented in Table 1, and correlations between the measures are in Fig. S1 (for descriptive statistics before imputation see Table S1). The outcomes of the trained SOM are presented in Fig. 1, showing how the SOM represents the values for each weight vector (i.e., the weights that correspond to each individual executive function measure) across the grid of nodes. Each panel shows the distribution of weights for a different executive function measure. Data-driven consensus clustering applied to the SOM suggested that a three-cluster solution was optimal (the map separated by cluster can be seen in Fig. 1).

Table 1 – Norm-referenced age-corrected means and standard deviations for the whole sample across all measures used in the self-organising map and clustering after imputation.

	Mean	SD
AWMA: Digit recall	91.62	15.59
AWMA: Backward digit recall	91.39	11.46
AWMA: Dot matrix	90.02	15.60
AWMA: Mr X	95.90	14.43
DKEFS: Trails: number-letter sequencing	6.14	3.44
DKEFS: Tower	9.41	2.28
TEA-Ch2: Vigil	8.26	3.20
TEA-Ch2: Reds, Blues, Bags and Shoes	7.69	3.36
BRIEF: Initiation	66.46	10.43
BRIEF: Inhibition	66.16	15.15
BRIEF: Monitoring	65.51	10.71
BRIEF: Shifting	68.24	14.52
BRIEF: Organisation	60.08	9.82
BRIEF: Planning	70.41	9.57
BRIEF: Working memory	73.28	9.60
BRIEF: Emotional control	64.60	13.30

Note. AWMA = Automated Working Memory Assessment, Standardised scores ($M = 100$, $SD = 15$); DKEFS = Delis-Kaplan Executive Function Battery, Scaled scores ($M = 10$, $SD = 3$); TEA-Ch2 = Test of Everyday Attention for Children 2 Scaled scores ($M = 10$, $SD = 3$); BRIEF = Behaviour Rating Inventory of Executive Function, T-scores ($M = 50$, $SD = 10$). All norm-referenced scores take age into account, TEA-Ch2 and BRIEF also take sex into account.

3.2. Cluster characterisation

The group means for each of the three data-driven groups on the executive function assessments used in the SOM are shown in Fig. 2 (for exact p -values see Table S3). One group (Cluster 1, $N = 204$) was characterised by close to age-expected levels across the rating-based assessments, with fewer difficulties compared to the other two clusters. Their performance on the task-based measures was poorer than their ratings and below age-expected levels. Overall, they had better task-based performance compared to children in Cluster 2 and poorer performance than those in Cluster 3. Based on the discrepancy between their rating- and task-based performance, children in Cluster 1 are referred to as having an inconsistent profile of predominantly task-based difficulties.

A second group, Cluster 2 ($N = 187$), was characterised by widespread difficulties across both the rating- and task-based executive function assessments and was therefore labelled as having consistent difficulties. The third group, Cluster 3 ($N = 175$), included children with the opposite profile to those in Cluster 1. They had an inconsistent profile with close to age-expected levels of performance on the task-based measures with pronounced difficulties on the rating-based assessments that were indistinguishable from those of the children in Cluster 2 who had a profile of consistent difficulties. Children in Cluster 3 are referred to as having an inconsistent profile of predominantly rating-based difficulties. Taken together the three data-driven groups had more homogenous profiles across the sixteen executive function assessments relative to groupings based on the largest diagnostic categories represented in our cohort (see Homogeneity analysis in Supplementary Materials for details).

There were significant differences in the average age of children in each of the three Clusters ($F = 9.56$, $p < .01$, see Table 2): children in Cluster 1 who had an inconsistent profile of predominantly task-based difficulties were on average younger than both children in Cluster 2 who had a consistent profile of difficulties ($t = -3.50$, $p < .001$, mean difference = .71 years) and children in Cluster 3 who had predominantly rating-based difficulties ($t = -3.06$, $p < .002$, mean difference = .62 years). There was no significant difference in age between children in Clusters 2 and 3. To explore whether child sex was related to group membership, while accounting for the overrepresentation of boys in the sample as a whole, a series of chi-square tests were conducted to compare the sex distribution observed in each cluster to the sex distribution of the whole sample (Table 2). Girls were overrepresented in Cluster 1 and underrepresented in Cluster 3 (profile with predominantly rating-based difficulties). The distribution of boys and girls in Cluster 2 (consistent difficulties group) matched that of the whole sample, suggesting that boys and girls were equally likely to have this profile.

3.3. Cluster validation

To validate differences between the three clusters, their performance was compared across measures of learning, mental health, and dimensions of neurodiversity, and the distribution of children with diagnosed neurodevelopmental conditions in each cluster was compared to the distribution across the whole

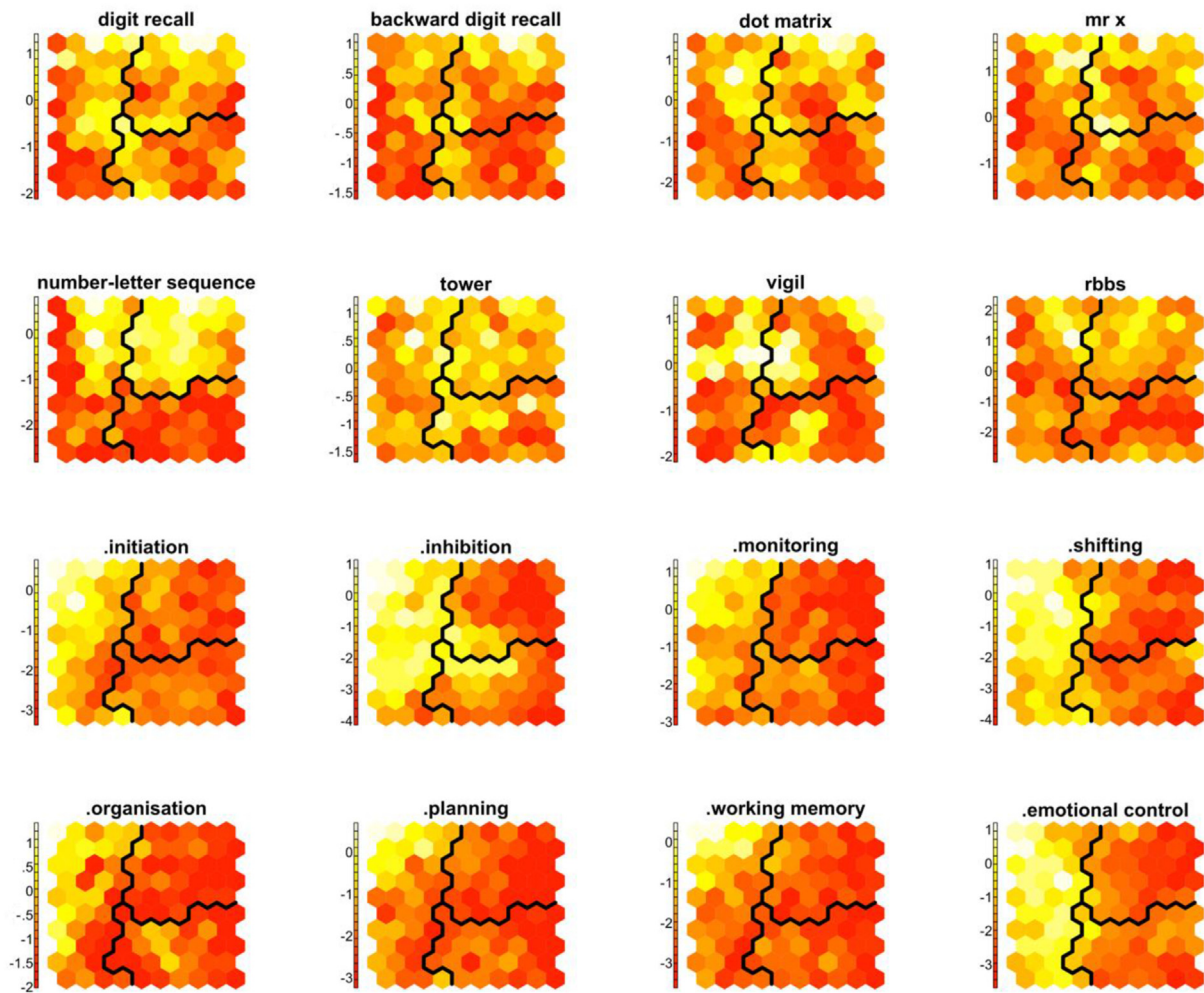


Fig. 1 – Weight distributions from the self-organizing map, split by assessment. For each assessment, the map depicts high weights (i.e., relatively stronger performance/ratings) as yellow hexagons and low weights (i.e., below age-expected performance/rating) as red hexagons. Boundaries represent the clustering solution: Cluster 1 is on the far left; Cluster 2 – bottom right, Cluster 3 – top right. Note. Automated Working Memory Assessment: digit recall, backward digit recall, dot matrix, mr x; Delis-Kaplan Executive Function Battery: trails number-letter sequencing & tower; Test of Everyday Attention for Children 2: vigil & reds, blues, bags and shoes; Behaviour Rating Inventory of Executive Function: inhibition; initiation, monitoring, shifting; organisation of materials, planning, working memory, emotional control. All scores are norm-referenced for age and BRIEF and TEA-Ch2 also take into account sex. They were converted to z-scores where scores of 0 are equivalent to age-typical levels, scores of 1 indicate one SD above the age-expected mean, and scores of –1 indicate one SD below the age-expected mean. Note that x- and y-axis carry no pre-determined meaning, rather the organisation is the output of the SOM algorithm.

sample. Children in Cluster 3 (inconsistent profile of predominantly rating-based difficulties) had significantly better performance across measures of reading and maths relative to children in the other two clusters (Fig. 3). Those in Cluster 2 (consistent difficulties profile) had significantly lower performance on the maths measure relative to children in Clusters 1 and 3, but their reading performance was indistinguishable from that of children in Cluster 1 who had predominantly task-based difficulties. For mental health, children in Cluster 3 (predominantly rating-based difficulties) had significantly elevated levels of anxiety and depression relative to children in the other two clusters; their scores were not significantly different from one another (see Fig. 3).

Children without any diagnoses were overrepresented in Cluster 1 ($p < .001$) but were represented in similar proportions in the other two clusters. Next, we looked at the largest diagnostic groups in our sample, observing that autistic participants and those with an ADHD diagnosis were underrepresented in Cluster 1, the group with predominately task-based difficulties (Table 2) but were represented in similar proportions in the other two clusters. It should be noted that 41 % of autistic participants also had an ADHD diagnosis, meaning separate conclusions cannot be drawn for the two conditions. Children with a dyslexia diagnosis were present in each cluster, and there was no relationship between diagnostic status for dyslexia and group membership (Table 2, p -s > .05). A

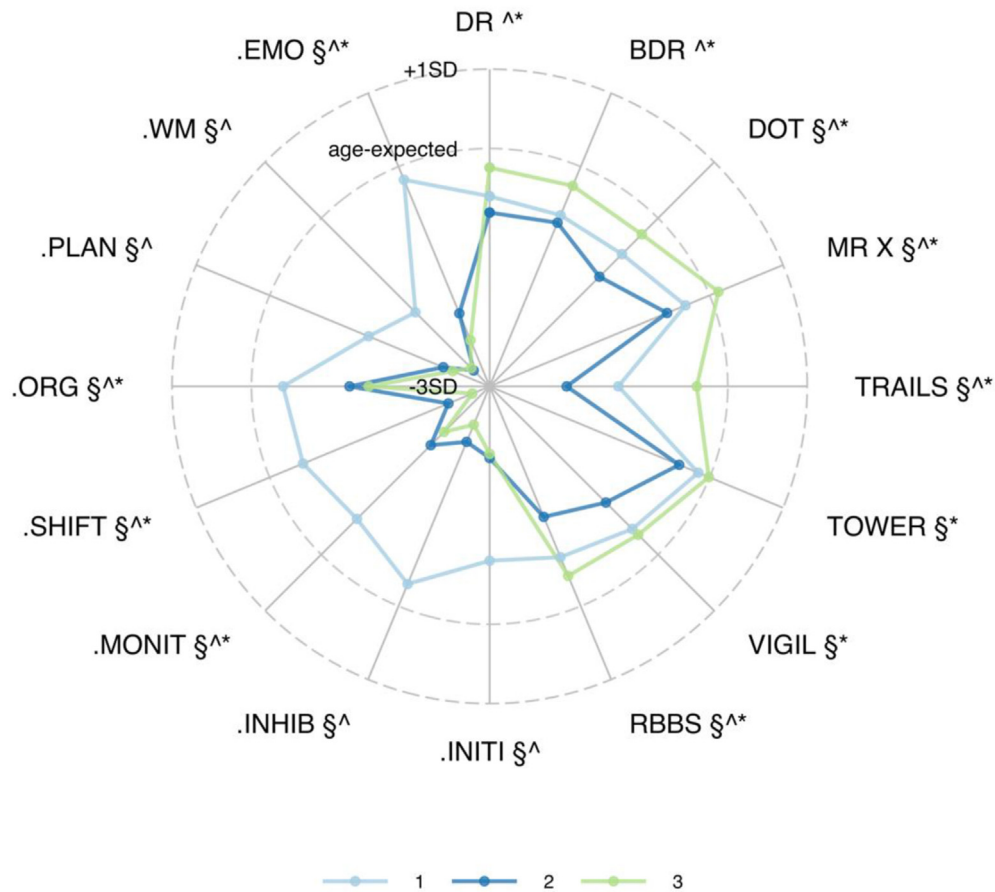


Fig. 2 – Executive function scores across all assessments used in the self-organising map for each one of the data-driven groups. Note. All age-referenced scores are converted to z-score, where scores of zero are equivalent to age-typical levels, scores of 1 indicate one standard deviation above the age-expected mean, and scores of -1 indicate one standard deviation below the age-expected mean. p -values are adjusted using false discovery rate correction. DR = AWMA Digit recall, BRD = AWMA Backward digit recall, DOT = AWMA Dot matrix, MR X = AWMA Mr X, TRAILS = DKEFS number-letter sequencing, TOWER = DKEFS Tower, VIGIL = TEA-Ch2 Vigil, RBBS = TEA-Ch2 Reds, Blues, Bags and Shoes, .INITI = BRIEF Initiation, INHIB = BRIEF Inhibition, .MONIT = BRIEF Monitoring, .SHIFT = BRIEF Shifting, .ORG = BRIEF Organisation, .PLAN = BRIEF Planning, .WM = BRIEF Working Memory, .EMO = BRIEF Emotional Control; AWMA = Automated Working Memory Assessment, DKEFS = Delis-Kaplan Executive Function Battery, TEA-Ch2 = Test of Everyday Attention for Children 2; BRIEF = Behaviour Rating Inventory of Executive Function. $\$ = p_{\text{fdr-corrected}} (1 \text{ vs } 2) < .05$; $^{\wedge} = p_{\text{fdr-corrected}} (1 \text{ vs } 3) < .05$; $^* = p_{\text{fdr-corrected}} (2 \text{ vs } 3) < .05$.

similar pattern emerged across dimensions associated with neurodevelopmental conditions. Across the five dimensional measures (inattention, hyperactivity/impulsivity, general communication skills, social communication and interests),

children in Cluster 1 (predominantly task-based difficulties) had significantly fewer difficulties than those in Clusters 2 and 3, who did not differ significantly from one another. In other words, there was no clear one-to-one correspondence between

Table 2 – Average age and distribution of sexes and different diagnoses across data-driven groups.

Cluster	Age	Sex			ADHD			Autism			Dyslexia		
		N_{boys}	χ^2	p	$N_{\text{diagnosed}}$	χ^2	p	$N_{\text{diagnosed}}$	χ^2	p	$N_{\text{diagnosed}}$	χ^2	p
1, $N = 204$	10.1	121	5.7	.035	16	39.7	0	3	12.1	.002	21	1.1	.4
2, $N = 187$	10.8	124	.1	.883	70	9.1	.008	21	2.4	.181	15	.0	.889
3, $N = 175$	10.7	135	7.9	.012	70	13.6	.001	22	4.6	.054	11	.9	.4

Note. Comparisons are based on chi-square tests, contrasting the overall proportion of boys and girls/diagnoses in the whole-referred sample to the proportion observed in each cluster. All p -values are adjusted using a false-discovery rate correction applied across all comparisons shown in this table. Note that diagnostic categories are not exclusive in this sample, and many children had multiple diagnoses.

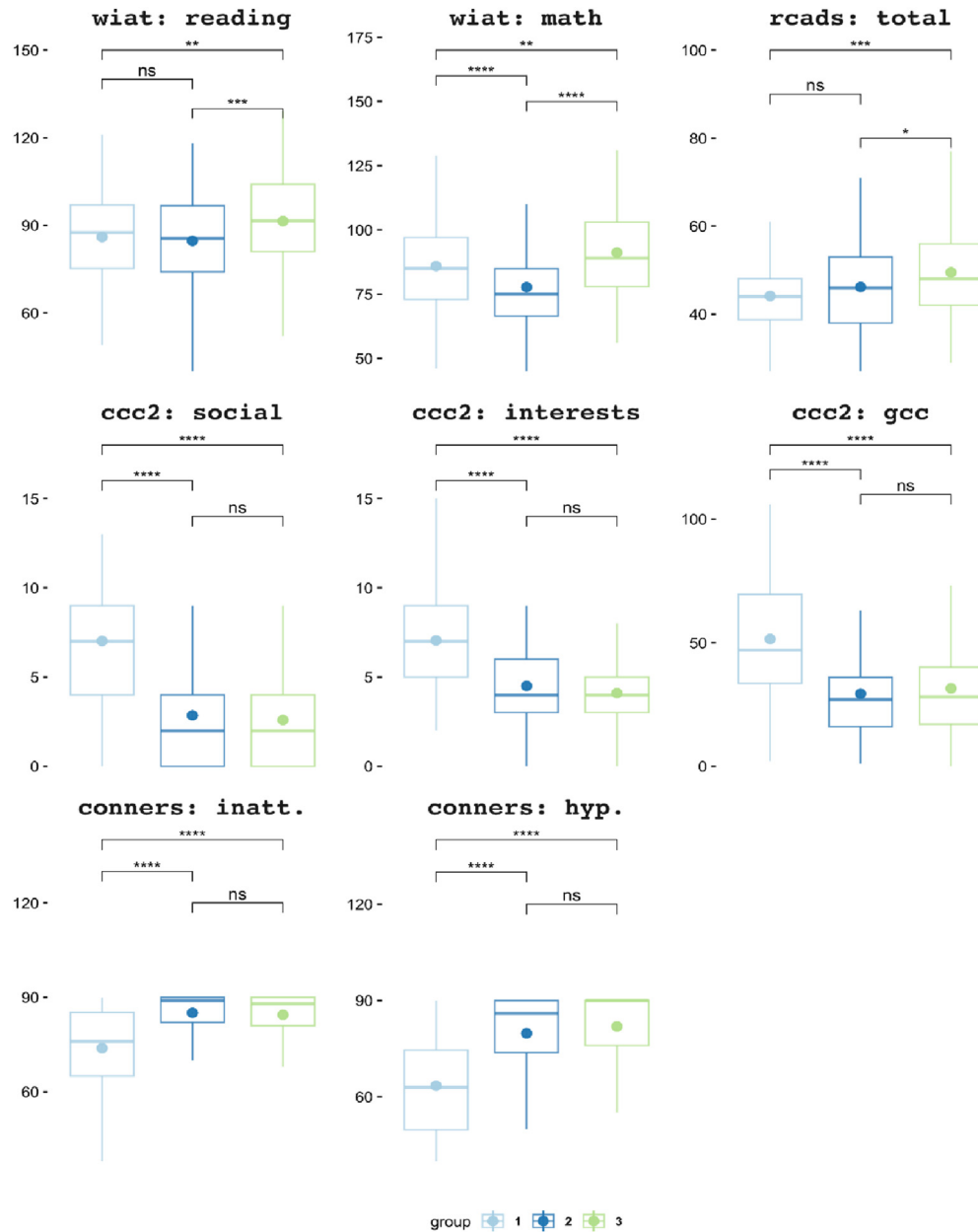


Fig. 3 – Results of pairwise comparisons of the data-driven groups across measures of learning, self-rated mental health, and parent-rated neurodevelopmental diagnostic dimensions. All measures are age-corrected. Note. WIAT = Wechsler Individual Achievement Test II: reading = Word Reading & math = numerical operations, Standardised scores, higher scores indicate relatively better performance; RCADS = The Revised Child Anxiety and Depression Scale (Child-version) total difficulties score, T-score, higher score indicate more self-reported difficulties; CCC2 = Children Communication Checklist-2: gcc = general communication composite, social = social relations, Scaled score, higher scores indicate relatively better skills; Conners = Conners-3: inattention and hyperactivity/impulsivity, T-scores, higher scores indicate relatively more difficulties; All p -values are adjusted using false discovery rate correction. $*p < .05$, $p < .01$, $***p < .001$, $****p < .0001$.**

data-driven group membership and neurodevelopmental diagnostic dimensions. Instead, relative to the group with primarily task-based difficulties (Cluster 1), the groups with consistent and predominantly rating-based difficulties (Clusters 2 and 3) included more children with diagnosed neurodevelopmental conditions and experienced more neurodevelopmental difficulties on dimensional measures.

3.4. Neuroimaging comparisons

At the global level (Fig. 4), children with consistent difficulties across rating- and task-based measures (Cluster 2) and those with mainly rating-based difficulties (Cluster 3) had significantly lower global connectome efficiency compared to the non-referred group. The three data-driven groups did not

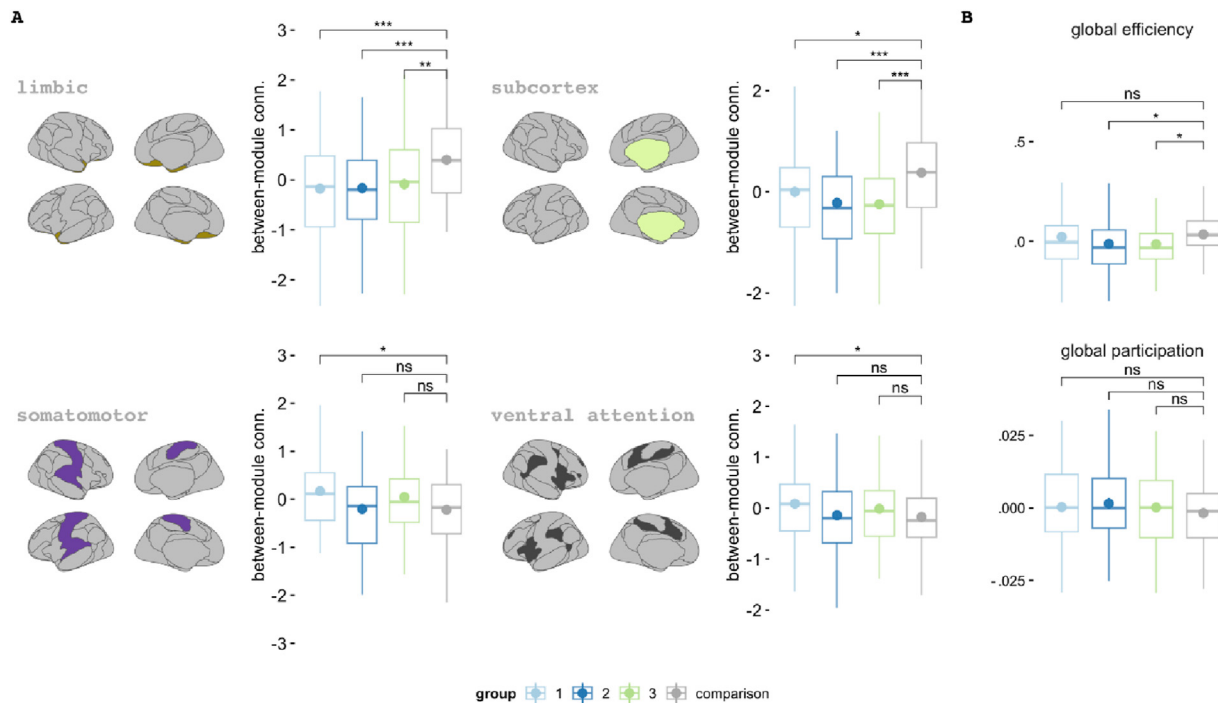


Fig. 4 – Group differences in modular and global organisation between each one of the three data-driven groups and the comparison group. Note. Panel A shows modules which showed different between-module strength from the comparison sample in at least one of the data-driven groups. Panel B shows group comparisons of global connectome metrics efficiency and participation coefficient. All metrics are age and motion-corrected. All p -values are corrected using false-discovery rate adjustment. $* < .05$, $ < .01$, $*** < .001$.**

differ significantly from the non-referred comparison group in average connectome participation coefficient. Looking at modular organisation, Clusters 2 (consistent difficulties) and 3 (predominantly rating-based difficulties) had similar patterns of neural organisation that differentiated them from the non-referred comparison group: they both showed reduced modular strength in the limbic network and the subcortex (Fig. S2). The group with primarily task-based difficulties (Cluster 1) showed the same pattern of reduced modular strength in the limbic network and the subcortex but was also characterised by increased strength in somatomotor and ventral attention modules relative to the comparison sample (Fig. S2). Follow-up analysis suggested that all these differences were driven by differences in between – rather than within – modular connection strength (Fig. 4).

Next, we identified 22 connector hubs that included regions within middle frontal gyrus, orbital gyrus, precentral gyrus, insula, cingulate gyrus, and the subcortex. The cross-validated balanced error rate across all models explored ($k = 1:8$) suggested that the most accurate PLSDA model was the one with two components. The bootstrapped hub loadings onto the PLSDA components, which had 95 % confidence intervals that did not cross zero, are shown in Fig. 5. For the first component, robust loadings included regions mostly within the insula, as well as within the dorsolateral prefrontal cortex, superior temporal gyrus, and the basal ganglia. Permutation analyses suggested that relative to the non-referred comparison group, children with an inconsistent profile of predominantly task-

based difficulties (Cluster 1) scored significantly lower on this component ($p = .003$). For the second component, all loadings that were shown to be robust in the bootstrapping procedure were frontal and parietal subregions. The second component distinguished participants with a profile of consistent difficulties across the executive function measures (Cluster 2) from the non-referred comparison group ($p = .02$), with component scores significantly higher in the comparison group.

4. Discussion

This study used an unsupervised machine learning approach to map profiles of executive function among a large neurodivergent sample of children. Self-organising maps were used to represent the profiles, which were subsequently carved into clusters representing homogeneous groups of children with similar profiles of executive function. Three clusters were identified, each with a distinct profile of executive function. The clusters could also be distinguished on measures of learning and mental health, but they cut across traditional neurodevelopmental diagnostic taxonomies: for the most part, they did not correspond to diagnostic categories or to dimensions of neurodiversity associated with common conditions such as ADHD or autism.

Three profiles of executive function were identified. One was characterised by difficulties that were consistent across the task- and rating-based measures of executive function

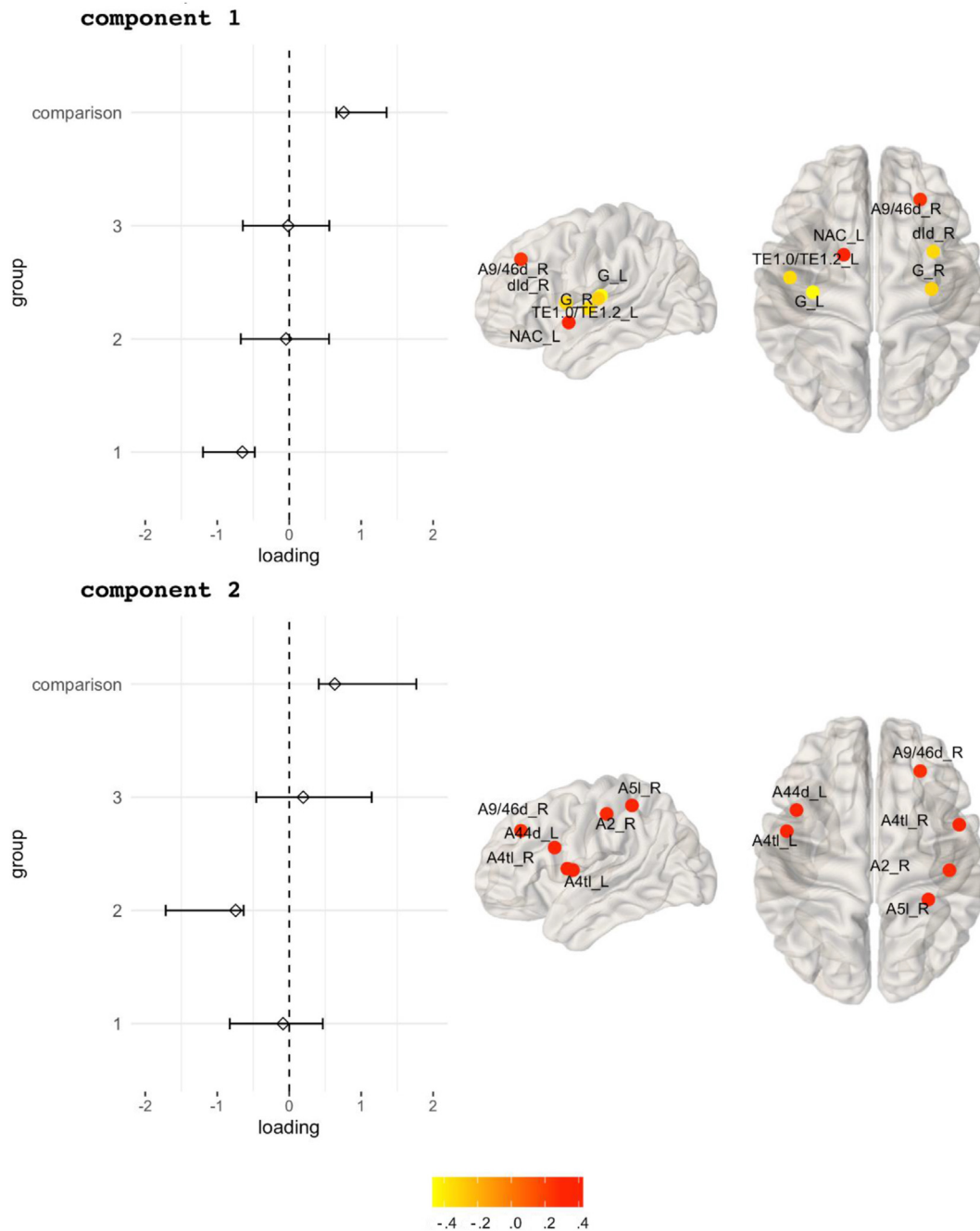


Fig. 5 – Bootstrapped component loadings of two components identified in the partial least squares discriminant analysis. Note. The forest plots show the group loadings for both components (95 % confidence intervals based on 1000 bootstraps). Displayed on the left side are connector hub loadings on both components. Only hub loadings that had 95 % confidence intervals that did not cross zero are displayed.

(Cluster 2). The other two profiles were inconsistent across the two measurement types: children in Cluster 1 had close to age-expected levels on the rating-based measures but had difficulties on the performance-based tasks, while children in Cluster 3 had the opposite profile of close to age-expected performance on the task-based measures with pronounced difficulties on the rating-based assessments. In total, more than half the sample had an inconsistent profile. This

provides further evidence that the two types of assessment (ratings and tasks) are dissociable, consistent with previous studies that have used latent variable or predictive validity approaches (e.g., Gerst et al., 2017; Nin et al., 2022; Soto et al., 2020; Tamm & Peugh, 2019; Toplak et al., 2013). The finding that a substantial proportion of our transdiagnostic neuro-divergent sample showed little overlap between performance and rating-based measures is also consistent with previous

neurodevelopmental literature, which has shown that the two domains of executive function can be selectively affected within the boundaries of a single neurodevelopmental condition (e.g., those with ADHD, [Biederman et al., 2008](#)). Overall, the current findings support the idea that the two types of measurement provide non-redundant information and should not be used interchangeably in the context of neurodevelopment.

4.1. Cluster comparisons: behaviour

Children in Cluster 1, who were characterised by difficulties on the cognitive performance-based tasks with near to age-typical everyday executive function behaviours, were significantly younger than children in other two clusters, were less likely to have a diagnosed neurodevelopmental condition, and had fewer difficulties on dimensions of neurodiversity associated with ADHD, developmental language disorder and autism than children in the other two clusters. Girls were over-represented in this cluster. Children in this cluster performed more poorly on both learning measures than children in Cluster 3, who had the opposite profile of better task-than rating-based performance, and their maths scores were equivalent to those in Cluster 2 who had a consistent but poor executive function profile.

The broad profile of children in Cluster 1 suggests that diagnoses are less common among younger children, girls, and those with everyday behaviours that appear to be age-typical, despite poor performance on cognitive task-based measures of executive function and learning. This might reflect diagnostic referral systems that rely primarily on observations of behaviour rather than performance on cognitive tests (the everyday behaviours of this group were almost age-typical). Currently, in England, referrals for diagnoses require an adult to notice a difficulty, and while diagnosis is not made solely based on a rating scale, the diagnostic process itself involves direct observation of behaviour in different settings ([National Institute for Health and Care Excellence, 2011, 2019](#)). The nature of the executive function difficulties of this group, which were likely less overt (i.e., not easily observed through outward behaviours) may have meant their difficulties were not as easily detected by parents and professionals, and therefore that fewer children were referred for diagnostic assessment/received a diagnosis. The lower diagnostic prevalence in this group may also be related to their age. This group was the youngest of all three, and considering that conditions such as ADHD are typically diagnosed at age 7 or above ([American Psychiatric Association, 2013](#); [Hoang et al., 2019](#); [Root et al., 2019](#)), it may be that diagnostic rates increase in this group as they age. Lower rates of diagnoses among this group could also reflect the overrepresentation of girls in this group – there are documented differences in the outward characteristics of neurodivergent girls and boys: girls are more likely to use behavioural camouflaging strategies, appearing more able from others' perspectives ([Dean et al., 2017](#); [Hiller et al., 2014](#); [Hull et al., 2020](#)), meaning they are less likely to receive a diagnosis ([Dhuey & Lipscomb, 2010](#); [Lockwood Estrin et al., 2021](#)).

Patterns of difficulties captured by parent/carer ratings on dimensions of neurodiversity associated with neurodevelopmental difficulties (e.g. ratings of inattention,

restricted social interests etc.) reflected the same pattern as the diagnostic categories: fewer difficulties were present in children in this cluster relative to those in the other clusters. This could again reflect masking/camouflaging, as well as socially constructed gender-biased or stereotypical views of boys as being disruptive (e.g., [Sciutto et al., 2004](#)). Together these findings underscore the need to include cognitive task-based assessments and female-representative characteristics in systems designed to identify children's additional needs (see [Guy et al., 2022](#) further discussion of this issue).

Children with the opposite profile of executive function, those in Cluster 3 with close to age-typical task-based performance and greater difficulties on the rating-based assessments, had significantly better performance on the learning measures than children in the other two clusters. This is consistent with a vast literature showing strong associations between performance on laboratory-based assessments of executive function and learning ([Bull et al., 2008](#); [Holmes et al., 2021](#); [St Clair-Thompson & Gathercole, 2006](#)). This group also self-reported more internalising difficulties relative to the other two clusters, potentially suggesting a link between higher levels of executive function rating difficulties and mental health. Notably, this could not be entirely attributed to rater biases, given that mental health was self-rated and executive function ratings were provided by parents. The relationship between mental health difficulties and a profile of inconsistent and predominantly rating-based difficulties has previously been observed using different methods in this cohort ([Williams et al., 2022](#)). Others have suggested that rating-based difficulties occurring in the absence of any task-based performance problems arise through emotional rather than cognitive mechanisms ([Vaidya et al., 2020](#); [Williams et al., 2022](#)), but longitudinal and/or causal evidence is needed to test this hypothesis.

Finally, those with a consistent profile of executive function difficulties had below age-expected performance across both types of executive function assessments. Relative to the other clusters, they had similar ratings of executive function difficulties to children in Cluster 3, but significantly poorer performance on the task-based assessments than children in both other clusters. Their reading and mental health ratings were equivalent to those in Cluster 1, but their maths performance was significantly poorer. They generally showed both commonalities and differences with children in the other two clusters, which were not entirely attributable to measurement type (e.g., even though their rating-based executive function difficulties were equivalent to those in Cluster 3, they had significantly different mental health ratings).

4.2. Cluster comparisons: neuroimaging

We observed both shared and cluster-specific patterns of neural white matter organisation differentiating data-driven groups from the comparison sample of children who had not been referred for attention, learning, and/or memory difficulties. Children with either a consistent profile of executive difficulties (Cluster 2) and those with predominantly task-based difficulties (Cluster 3) had decreased global connectome efficiency relative to the comparison sample, but no group differences were observed in global participation. This was

unexpected given previous reports of an association between modular segregation and executive function (Baum et al., 2017).

Looking at modular organisation, all three data-driven groups showed reduced connection strength within the limbic network and the subcortex relative to the non-referred comparison group, with follow-up analyses showing that differences were driven by reduced inter-module connections. This is consistent with the idea that limbic and subcortical areas play a role in goal-directed cognition and that differences in this circuitry are related to executive functioning in neurodevelopmental conditions (e.g., Arnsten & Rubia, 2012). The group with predominantly task-based difficulties (Cluster 1) also showed increased inter-module connection strength in the somatomotor and ventral attention modules, a pattern that was also observed in the PLSDA: the component of connector hubs differentiating this group from the non-referred comparison group involved multiple insular regions of the somatomotor and ventral attention modules. Somatomotor and ventral attention networks undergo substantial restructuring over the course of development, becoming more structurally segregated with age (Baum et al., 2017; Grayson & Fair, 2017). Observing differences in this network among children with predominantly task-based difficulties is consistent with their implicated role in task-based executive function (Reineberg et al., 2015). Additionally, several of the hubs which loaded robustly on the PLSDA component that differentiated this group from the non-referred sample shared perception as a behavioural metadata label within the BrainMap Database (<http://www.brainmap.org/taxonomy>).

Children with a profile of consistent difficulties across the task- and rating-based measures (Cluster 2) were uniquely distinguished from the non-referred comparison group through a component of frontal and parietal connector hubs identified through the PLSDA. This is consistent with decades of research that has established a role for frontal and parietal regions in supporting executive function (see Friedman & Robbins, 2022). Exploratory cross-checking of the behavioural metadata labels for the hubs robustly loading on this component within the BrainMap Database suggested that many of them were functionally implicated in action execution.

Finally, considering other similarities, the two components identified by the PLSDA that differentiated the two groups with task-based executive function difficulties (Cluster 1 and 2) from the non-referred comparison sample had a single shared robust loading. This region, located in the right dorsal area 9/46 within the middle frontal gyrus, is functionally associated with task-based executive function performance on measures of working memory and attention (e.g., Jung et al., 2022).

4.3. Implications & future directions

The current findings speak to the need to incorporate both performance- and rating-based assessments of executive function in the diagnostic process or when tailoring support and accommodations for neurodivergent children. In the research domain, we advocate that researchers explicitly specify their choice and rationale for the aspect of executive function they are interested in, or better still, consider both in combination. We particularly encourage further research into

the developmental origins and longitudinal trajectories of the profiles observed here, which could provide important insights into the mechanisms that distinguish the two aspects of executive function.

4.4. Limitations

The measurement types used were limited by those available in the cohort data. As such, it is possible, for example, that stronger links between the everyday manifestations of executive function difficulties captured by the rating scales and learning could have been detected if we did not have to rely on measures of learning administered under optimal testing conditions, but instead had access to school-based measures of academic achievement that reflect learning and assessment in everyday conditions. The age-heterogenous nature of the cohort meant that age was controlled for in all analyses (i.e., all scores were age-corrected), which limited our ability to explore age-related differentiation between different executive skills. We also note that the CALM cohort is drawn from the school population of Southeast England, which may limit the extent to which these findings generalise to populations not well-represented in this part of the country. Methodologically, the combination of a continuous multidimensional mapping method with a data-driven clustering algorithm means that inevitably some children will sit close to the cluster boundaries within the map. This issue is common to any attempt to parse out heterogeneity by applying categorical boundaries to any type of continuous space. In terms of the neuroimaging results, it is important to acknowledge that the use of multimodal neuroimaging and larger samples may reveal further and equally important differences across groups, which we consider an important future direction. Finally, we note that exploratory data-driven methods could sometimes “overfit” the data (i.e., capture noise and irrelevant details), which may limit generalisability to unseen data. We, therefore, encourage further efforts to replicate and extend these findings to better understand the diversity of executive function profiles among neurodivergent individuals.

5. Conclusion

In summary, we used a machine learning approach to map the executive function profiles of large neurodevelopmentally neurodivergent sample. We identified three distinct profiles that were validated by differences across measures of learning, mental health, and neural white matter organisation. The profiles did not clearly map onto neurodevelopmental diagnostic categories, but we observed that children with an inconsistent profile of primarily task-based difficulties were less likely to have a diagnosis and more likely to be girls. These data add to the growing evidence base for two dissociable aspects of executive function: one that reflects state-like efficiency of cognitive skills and can be measured by cognitive tasks (Miyake et al., 2000), and another that reflects the trait-like application of cognitive control in everyday contexts and can be measured by rating scales (Doebel, 2020). The majority of our neurodivergent sample had inconsistent profiles across the two domains, suggesting that

researchers and practitioners should not use these two types of assessments interchangeably in the context of neurodevelopment. They should instead be used together in a complementary manner.

Data and code availability statement

The CALM dataset is freely available to the research community through a managed open-access portal, subject to terms and conditions ensuring ethical and responsible data use. For details, please visit <https://portal.ide-cam.org.uk/overview/1158>. The analysis code is available online <https://github.com/sylmma/NeurodiversityInEF>.

Open practices

The study in this article earned Open Data Badge for transparent practices. The analysis code are available at (<https://github.com/sylmma/NeurodiversityInEF>) and the raw data are available at <https://portal.ide-cam.org.uk/overview/1158>.

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Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cortex.2023.11.021>.

REFERENCES

- Alloway, T. P. (2007). *Automated Working Memory Assessment: Manual*. Pearson.
- Alloway, T. P., Gathercole, S. E., Kirkwood, H., & Elliott, J. (2009). The working memory rating scale: A classroom-based behavioral assessment of working memory. *Learning and Individual Differences*, 19(2), 242–245. <https://doi.org/10.1016/j.lindif.2008.10.003>
- Ameis, S. H., Lerch, J. P., Taylor, M. J., Lee, W., Viviano, J. D., Pipitone, J., Nazeri, A., Croarkin, P. E., Voineskos, A. N., Lai, M. C., Crosbie, J., Brian, J., Soreni, N., Schachar, R., Szatmari, P., Arnold, P. D., & Anagnostou, E. (2016). A diffusion tensor imaging study in children with ADHD, autism spectrum disorder, OCD, and matched controls: Distinct and non-distinct white matter disruption and dimensional brain-behavior relationships. *American Journal of Psychiatry*, 173(12), 1213–1222. <https://doi.org/10.1176/appi.ajp.2016.15111435>
- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders (DSM-5)*. American Psychiatric Pub.
- Armsten, A. F. T., & Rubia, K. (2012). Neurobiological circuits regulating attention, cognitive control, motivation, and emotion: Disruptions in neurodevelopmental psychiatric disorders. *Journal of the American Academy of Child and Adolescent Psychiatry*, 51(4), 356–367. <https://doi.org/10.1016/j.jaac.2012.01.008>
- Astle, D. E., Holmes, J., Kievit, R., & Gathercole, S. E. (2022). Annual Research Review: The transdiagnostic revolution in neurodevelopmental disorders. *Journal of Child Psychology and Psychiatry*, 63(4), 397–417. <https://doi.org/10.1111/jcpp.13481>
- Baggetta, P., & Alexander, P. A. (2016). Conceptualization and operationalization of executive function. *Mind, Brain, and Education*, 10(1), 10–33. <https://doi.org/10.1111/mbe.12100>
- Barkley, R. A., & Murphy, K. R. (2010). Impairment in occupational functioning and adult ADHD: The predictive utility of executive function (EF) ratings versus EF tests. *Archives of Clinical Neuropsychology*, 25(3), 157–173. <https://doi.org/10.1093/arclin/acq014>
- Barkley, R. A., & Murphy, K. R. (2011). The nature of executive function (EF) deficits in daily life activities in adults with ADHD and their relationship to performance on EF tests. *Journal of Psychopathology and Behavioral Assessment*, 33(2), 137–158. <https://doi.org/10.1007/s10862-011-9217-x>
- Bathelt, J., Holmes, J., Astle, D. E., Holmes, J., Gathercole, S., Astle, D., Manly, T., & Kievit, R. (2018). Data-driven subtyping of executive function-related behavioral problems in children. *Journal of the American Academy of Child and Adolescent Psychiatry*, 57(4), 252–262.e4. <https://doi.org/10.1016/j.jaac.2018.01.014>
- Bathelt, J., Vignoles, A., & Astle, D. E. (2021). Just a phase? Mapping the transition of behavioural problems from childhood to adolescence. *Social Psychiatry and Psychiatric Epidemiology*, 56(5), 821–836. <https://doi.org/10.1007/s00127-020-02014-4>
- Baum, G. L., Ciric, R., Roalf, D. R., Betzel, R. F., Moore, T. M., Shinohara, R. T., Kahn, A. E., Vandekar, S. N., Rupert, P. E., Quarmley, M., Cook, P. A., Elliott, M. A., Ruparel, K., Gur, R. E., Gur, R. C., Bassett, D. S., & Satterthwaite, T. D. (2017). Modular segregation of structural brain networks supports the development of executive function in youth. *Current Biology*, 27(11), 1561–1572.e8. <https://doi.org/10.1016/j.cub.2017.04.051>
- Beare, R., Adamson, C., Bellgrove, M. A., Vilgis, V., Vance, A., Seal, M. L., & Silk, T. J. (2017). Altered structural connectivity in ADHD: A network based analysis. *Brain Imaging and Behavior*, 11(3), 846–858. <https://doi.org/10.1007/s11682-016-9559-9>
- Benallie, K. J., McClain, M. B., Bakner, K. E., Roanhorse, T., & Ha, J. (2021). Executive functioning in children with ASD + ADHD and ASD + ID: A systematic review. *Research in Autism*

- Spectrum Disorders, 86, Article 101807. <https://doi.org/10.1016/J.RASD.2021.101807>
- Biederman, J., Petty, C. R., Fried, R., Black, S., Faneuil, A., Doyle, A. E., Seidman, L. J., & Faraone, S. V. (2008). Discordance between psychometric testing and questionnaire-based definitions of executive function deficits in individuals with ADHD. *Journal of Attention Disorders*, 12(1), 92–102. <https://doi.org/10.1177/1087054707305111>
- Bishop, D. V. M. (2003). *Children's Communication Checklist-2 (CCC-2)*. Pearson Assessment.
- Bloemen, A. J. P., Oldehinkel, A. J., Laceulle, O. M., Ormel, J., Rommelse, N. N. J., & Hartman, C. A. (2018). The association between executive functioning and psychopathology: General or specific? *Psychological Medicine*. <https://doi.org/10.1017/S0033291717003269>
- Brereton, R. G., & Lloyd, G. R. (2014). Partial least squares discriminant analysis: Taking the magic away. *Journal of Chemometrics*, 28(4), 213–225. <https://doi.org/10.1002/cem.2609>
- Bull, R., Espy, K. A., & Wiebe, S. A. (2008). Short-term memory, working memory, and executive functioning in preschoolers: Longitudinal predictors of mathematical achievement at age 7 years. *Developmental Neuropsychology*, 33(3), 205–228. <https://doi.org/10.1080/87565640801982312>
- Carlson, S. M., Moses, L. J., & Claxton, L. J. (2004). Individual differences in executive functioning and theory of mind: An investigation of inhibitory control and planning ability. *Journal of Experimental Child Psychology*, 87(4), 299–319. <https://doi.org/10.1016/j.jecp.2004.01.002>
- Charrad, M., Ghazzali, N., Boiteau, V., & Niknafs, A. (2014). NbClust: An R package for determining the relevant number of clusters in a data set. *Journal of Statistical Software*, 61, 1–36. <https://doi.org/10.18637/jss.v061.i06>
- Chorpita, B. F., Moffitt, C. E., & Gray, J. (2005). Psychometric properties of the Revised Child Anxiety and Depression Scale in a clinical sample. *Behaviour Research and Therapy*, 43(3), 309–322. <https://doi.org/10.1016/J.BRAT.2004.02.004>
- Conners, C. K. (1992). *Conners' Continuous Performance Test user's manual*. Multi-Health System.
- Conners, C. K. (2008). *Conners 3rd edition (Conners 3) (Third Edit)*. Ontario, Canada: Multi-Health Systems Toronto.
- Dalmajer, E. S., Nord, C. L., & Astle, D. E. (2022). Statistical power for cluster analysis. *BMC Bioinformatics*, 23(1), 1–28. <https://doi.org/10.1186/s12859-022-04675-1>
- Dang, J., King, K. M., & Inzlicht, M. (2020). Why are self-report and behavioral measures weakly correlated? *Trends in Cognitive Sciences*, 24(4), 267–269. <https://doi.org/10.1016/j.tics.2020.01.007>
- Dean, M., Harwood, R., & Kasari, C. (2017). The art of camouflage: Gender differences in the social behaviors of girls and boys with autism spectrum disorder. *Autism: the International Journal of Research and Practice*, 21(6), 678–689. <https://doi.org/10.1177/1362361316671845>
- Delis, D. C., Kaplan, E., & Kramer, J. H. (2001). *Delis-Kaplan executive function system (D-KEFS)*. The Psychological Corporation.
- Deodhar, A. V., & Bertenthal, B. I. (2023). How attention factors into executive function in preschool children. *Frontiers in Psychology*, 14, Article 1146101. <https://doi.org/10.3389/fpsyg.2023.1146101>
- Dhuey, E., & Lipscomb, S. (2010). Disabled or young? Relative age and special education diagnoses in schools. *Economics of Education Review*, 29(5), 857–872. <https://doi.org/10.1016/j.econedurev.2010.03.006>
- Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, 64(1), 135–168. <https://doi.org/10.1146/annurev-psych-113011-143750>
- Doebel, S. (2020). Rethinking executive function and its development. *Perspectives on Psychological Science*, 15(4), 942–956. <https://doi.org/10.1177/1745691620904771>
- Doebel, S., & Müller, U. (2023). The future of research on executive function and its development: An introduction to the special issue. *Journal of Cognition and Development*, 24(2), 161–171. <https://doi.org/10.1080/15248372.2023.2188946>
- Eisenberg, I. W., Bissett, P. G., Zeynep Enkavi, A., Li, J., MacKinnon, D. P., Marsch, L. A., & Poldrack, R. A. (2019). Uncovering the structure of self-regulation through data-driven ontology discovery. *Nature Communications*, 10(1). <https://doi.org/10.1038/s41467-019-10301-1>. Article 1.
- Ellingson, J. M., Corley, R., Hewitt, J. K., & Friedman, N. P. (2019). A prospective study of alcohol involvement and the dual-systems model of adolescent risk-taking during late adolescence and emerging adulthood. *Addiction*, 114(4), 653–661. <https://doi.org/10.1111/add.14489>
- Fan, L., Li, H., Zhuo, J., Zhang, Y., Wang, J., Chen, L., Yang, Z., Chu, C., Xie, S., Laird, A. R., Fox, P. T., Eickhoff, S. B., Yu, C., & Jiang, T. (2016). The human Brainnetome atlas: A new brain atlas based on connectational architecture. *Cerebral Cortex*, 26(8), 3508–3526. <https://doi.org/10.1093/cercor/bhw157>
- Friedman, N. P., & Gustavson, D. E. (2022). Do rating and task measures of control abilities assess the same thing? *Current Directions in Psychological Science*, 31(3), 262–271. <https://doi.org/10.1177/09637214221091824>
- Friedman, N. P., Hatoum, A. S., Gustavson, D. E., Corley, R. P., Hewitt, J. K., & Young, S. E. (2020). Executive functions and impulsivity are genetically distinct and independently predict psychopathology: Results from two adult twin studies. *Clinical Psychological Science*, 8(3), 519–538. <https://doi.org/10.1177/2167702619898814>
- Friedman, N. P., & Miyake, A. (2017). Unity and diversity of executive functions: Individual differences as a window on cognitive structure. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, 86, 186–204. <https://doi.org/10.1016/J.CORTEX.2016.04.023>
- Friedman, N. P., & Robbins, T. W. (2022). The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology: Official Publication of the American College of Neuropsychopharmacology*, 47(1). <https://doi.org/10.1038/s41386-021-01132-0>. Article 1.
- Gómez-Pérez, M. M., Calero, M. D., Mata, S., & Molinero, C. (2016). Discrepancies between direct and indirect measures of interpersonal and neurocognitive skills in autism spectrum disorder children. *Journal of Clinical and Experimental Neuropsychology*, 38(8), 875–886. <https://doi.org/10.1080/13803395.2016.1170106>
- Garon, N., Bryson, S. E., & Smith, I. M. (2008). Executive function in preschoolers: A review using an integrative framework. *Psychological Bulletin*, 134(1), 31–60. <https://doi.org/10.1037/0033-2909.134.1.31>
- Gerst, E. H., Cirino, P. T., Fletcher, J. M., & Yoshida, H. (2017). Cognitive and behavioral rating measures of executive function as predictors of academic outcomes in children. *Child Neuropsychology: A Journal on Normal and Abnormal Development in Childhood and Adolescence*, 23(4), 381–407. <https://doi.org/10.1080/09297049.2015.1120860>
- Geurts, H. M., Verté, S., Oosterlaan, J., Roeyers, H., & Sergeant, J. A. (2004). How specific are executive functioning deficits in attention deficit hyperactivity disorder and autism? *Journal of Child Psychology and Psychiatry*, 45(4), 836–854. <https://doi.org/10.1111/j.1469-7610.2004.00276.x>
- Gioia, G. A., Isquith, P. K., Guy, S. C., & Kenworthy, L. (2000). *Behavior rating inventory of executive function: BRIEF*. Odessa, FL: Psychological Assessment Resources.
- Grayson, D. S., & Fair, D. A. (2017). Development of large-scale functional networks from birth to adulthood: A guide to the neuroimaging literature. *NeuroImage*, 160, 15–31. <https://doi.org/10.1016/j.neuroimage.2017.01.079>

- Gross, A. C., Deling, L. A., Wozniak, J. R., & Boys, C. J. (2015). Objective measures of executive functioning are highly discrepant with parent-report in fetal alcohol spectrum disorders. *Child Neuropsychology*, 21(4), 531–538. <https://doi.org/10.1080/09297049.2014.911271>
- Guy, J., Mareva, S., Franckel, G., Team, the C., & Holmes, J. (2022). Dimensions of cognition, behaviour, and mental health in struggling learners: A spotlight on girls. *JCPP Advances*, 2(4), Article e12082. <https://doi.org/10.1002/jcv2.12082>
- Hedge, C., Powell, G., & Sumner, P. (2018). The reliability paradox: Why robust cognitive tasks do not produce reliable individual differences. *Behavior Research Methods*, 50(3), 1166–1186. <https://doi.org/10.3758/s13428-017-0935-1>
- Hiller, R. M., Young, R. L., & Weber, N. (2014). Sex differences in autism spectrum disorder based on DSM-5 criteria: Evidence from clinician and teacher reporting. *Journal of Abnormal Child Psychology*, 42(8), 1381–1393. <https://doi.org/10.1007/s10802-014-9881-x>
- Hoang, U., James, A. C., Liyanage, H., Jones, S., Joy, M., Blair, M., Rigby, M., & de Lusignan, S. (2019). Determinants of inter-practice variation in ADHD diagnosis and stimulant prescribing: Cross-sectional database study of a national surveillance network. *BMJ Evidence-Based Medicine*, 24(4), 155–161. <https://doi.org/10.1136/bmjebm-2018-111133>
- Holmes, J., Bryant, A., & Gathercole, S. E. (2019). Protocol for a transdiagnostic study of children with problems of attention, learning and memory (CALM). *BMC Pediatrics*, 19(1), 10. <https://doi.org/10.1186/s12887-018-1385-3>
- Holmes, J., Guy, J., Kievit, R. A., Bryant, A., Mareva, S., CALM Team, Gathercole, S. E. (2021). Cognitive dimensions of learning in children with problems in attention, learning, and memory. *Journal of Educational Psychology*, 113(7), 1454–1480. <https://doi.org/10.1037/edu0000644>
- Holmes, J., Hilton, K. A., Place, M., Alloway, T. P., Elliott, J. G., & Gathercole, S. E. (2014). Children with low working memory and children with ADHD: Same or different? *Frontiers in Human Neuroscience*, 8, 976. <https://doi.org/10.3389/fnhum.2014.00976>
- Holochwost, S. J., Winebrake, D., Brown, E. D., Happaney, K. R., Wagner, N. J., & Mills-Koonce, W. R. (2023). An ecological systems perspective on individual differences in children's performance on measures of executive function. *Journal of Cognition and Development*, 24(2), 223–240. <https://doi.org/10.1080/15248372.2022.2160721>
- Hull, L., Petrides, K. V., & Mandy, W. (2020). The female autism phenotype and camouflaging: A narrative review. *Review Journal of Autism and Developmental Disorders*, 7(4), 306–317. <https://doi.org/10.1007/s40489-020-00197-9>
- Jones, J. S., the CALM Team, & Astle, D. E. (2021). A transdiagnostic data-driven study of children's behaviour and the functional connectome. *Developmental Cognitive Neuroscience*, 52, Article 101027. <https://doi.org/10.1016/j.DCN.2021.101027>
- Jung, J., Lambon Ralph, M. A., & Jackson, R. L. (2022). Subregions of DLPFC display graded yet distinct structural and functional connectivity. *The Journal of Neuroscience: the Official Journal of the Society for Neuroscience*, 42(15), 3241–3252. <https://doi.org/10.1523/JNEUROSCI.1216-21.2022>
- Koenis, M. M. G., Brouwer, R. M., van den Heuvel, M. P., Mandl, R. C. W., van Soelen, I. L. C., Kahn, R. S., Boomsma, D. I., & Hulshoff Pol, H. E. (2015). Development of the brain's structural network efficiency in early adolescence: A longitudinal DTI twin study. *Human Brain Mapping*, 36(12), 4938–4953. <https://doi.org/10.1002/hbm.22988>
- Krieger, V., & Amador-Campos, J. A. (2018). Assessment of executive function in ADHD adolescents: Contribution of performance tests and rating scales. *Child Neuropsychology*, 24(8), 1063–1087. <https://doi.org/10.1080/09297049.2017.1386781>
- Krishnan, A., Williams, L. J., McIntosh, A. R., & Abdi, H. (2011). Partial Least Squares (PLS) methods for neuroimaging: A tutorial and review. *NeuroImage*, 56(2), 455–475. <https://doi.org/10.1016/j.neuroimage.2010.07.034>
- Lockwood Estrin, G., Milner, V., Spain, D., Happé, F., & Colvert, E. (2021). Barriers to autism spectrum disorder diagnosis for young women and girls: A systematic review. *Review Journal of Autism and Developmental Disorders*, 8(4), 454–470. <https://doi.org/10.1007/s40489-020-00225-8>
- Loe, I. M., & Feldman, H. M. (2007). Academic and educational outcomes of children with ADHD. *Journal of Pediatric Psychology*, 32(6), 643–654. <https://doi.org/10.1093/jpepsy/jsl054>
- Malanchini, M., Engelhardt, L. E., Grotzinger, A. D., Harden, K. P., & Tucker-Drob, E. M. (2018). 'Same but different': Associations between multiple aspects of self-regulation, cognition, and academic abilities. *Journal of Personality and Social Psychology*. <https://doi.org/10.1037/pspp0000224>
- Manly, T., Anderson, V., Crawford, G., George, M., & Robertson, I. (2016). *Test of Everyday Attention for Children, Second Edition (TEA-Ch2)*. Pearson. <https://www.pearsonclinical.co.uk/store/ukassessments/en/adaptive-behavior/Test-of-Everyday-Attention-for-Children%2C-Second-Edition/p/P100009285.html>
- Mareva, S., Akarca, D., Team, T. C., & Holmes, J. (2023). Transdiagnostic profiles of behaviour and communication relate to academic and socioemotional functioning and neural white matter organisation. *Journal of Child Psychology and Psychiatry*, 64(2), 217–233. <https://doi.org/10.1111/jcpp.13685>
- Mareva, S., & Holmes, J. (2019). Transdiagnostic associations across communication, cognitive, and behavioural problems in a developmentally at-risk population: A network approach. *BMC Pediatrics*, 19(1). <https://doi.org/10.1186/s12887-019-1818-7>, 452–452.
- McClain, M. B., Golson, M. E., & Murphy, L. E. (2022). Executive functioning skills in early childhood children with autism, intellectual disability, and co-occurring autism and intellectual disability. *Research in Developmental Disabilities*, 122, Article 104169. <https://doi.org/10.1016/j.ridd.2021.104169>
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex 'Frontal Lobe' tasks: A latent variable analysis. *Cognitive Psychology*, 41(1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>
- National Institute of Health and Care Excellence. (2011). Recommendations: Autism spectrum disorder in under 19s: Recognition, referral and diagnosis. <https://www.nice.org.uk/guidance/cg128/chapter/Recommendations#recognising-children-and-young-people-with-possible-autism>
- National Institute of Health and Care Excellence. (2019). Recommendations: Attention deficit hyperactivity disorder: Diagnosis and management. <https://www.nice.org.uk/guidance/ng87/chapter/Recommendations#recognition-identification-and-referral>
- Nin, V., Delgado, H., Muniz-Terrera, G., & Carboni, A. (2022). Partial agreement between task and BRIEF-P-based EF measures depends on school socioeconomic status. *Developmental Science*, 25(5), Article e13241. <https://doi.org/10.1111/desc.13241>
- Peng, P., & Fuchs, D. (2016). A meta-analysis of working memory deficits in children with learning difficulties: Is there a difference between verbal domain and numerical domain? *Journal of Learning Disabilities*, 49(1), 3–20. <https://doi.org/10.1177/0022219414521667>
- Pezzoli, P., Parsons, S., Kievit, R. A., Astle, D. E., Huys, Q. J. M., Steinbeis, N., & Viding, E. (2023). Challenges and solutions to

- measurement of neurocognitive mechanisms in developmental settings. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, 0(0). <https://doi.org/10.1016/j.bpsc.2023.03.011>
- Reineberg, A. E., Andrews-Hanna, J. R., Depue, B. E., Friedman, N. P., & Banich, M. T. (2015). Resting-state networks predict individual differences in common and specific aspects of executive function. *NeuroImage*, 104, 69–78. <https://doi.org/10.1016/j.neuroimage.2014.09.045>
- Rohart, F., Gautier, B., Singh, A., & Cao, K.-A. L. (2017). mixOmics: An R package for 'omics feature selection and multiple data integration. *PLoS Computational Biology*, 13(11), Article e1005752. <https://doi.org/10.1371/journal.pcbi.1005752>
- Root, A., Brown, J. P., Forbes, H. J., Bhaskaran, K., Hayes, J., Smeeth, L., & Douglas, I. J. (2019). Association of relative age in the school year with diagnosis of intellectual disability, attention-deficit/hyperactivity disorder, and depression. *JAMA Pediatrics*, 173(11), 1068–1075. <https://doi.org/10.1001/jamapediatrics.2019.3194>
- Scituito, M. J., Nolfi, C. J., & Bluhm, C. (2004). Effects of child gender and symptom type on referrals for ADHD by elementary school teachers. *Journal of Emotional and Behavioral Disorders*, 12(4), 247–253. <https://doi.org/10.1177/10634266040120040501>
- Shallice, T. (1982). Specific impairments of planning. *Philosophical Transactions of the Royal Society of London, Biology*, 289, 199–209.
- Snyder, H. R., Friedman, N. P., & Hankin, B. L. (2021). Associations between task performance and self-report measures of cognitive control: Shared versus distinct abilities. *Assessment*, 28(4), 1080–1096. <https://doi.org/10.1177/1073191120965694>
- Soto, E. F., Irwin, L. N., Chan, E. S. M., Spiegel, J. A., & Kofler, M. J. (2021). Executive functions and writing skills in children with and without ADHD. *Neuropsychology*, 35, 792–808. <https://doi.org/10.1037/neu0000769>
- Soto, E. F., Kofler, M. J., Singh, L. J., Wells, E. L., Irwin, L. N., Groves, N. B., & Miller, C. E. (2020). Executive functioning rating scales: Ecologically valid or construct invalid? *Neuropsychology*, 34(6), 605–619. <https://doi.org/10.1037/neu0000681>
- Sporns, O., Honey, C., & Kötter, R. (2007). Identification and classification of hubs in brain networks. *PLoS One*, 2(10), e1049. <https://doi.org/10.1371/journal.pone.0001049>
- St Clair-Thompson, H. L., & Gathercole, S. E. (2006). Executive functions and achievements in school: Shifting, updating, inhibition, and working memory. *Quarterly Journal of Experimental Psychology* (2006), 59(4), 745–759. <https://doi.org/10.1080/17470210500162854>
- Stekhoven, D. J., & Bühlmann, P. (2012). Missforest-Non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1), 112–118. <https://doi.org/10.1093/bioinformatics/btr597>
- Strauss, E., Sherman, E. M. S., & Spreen, O. (2006). *A compendium of neuropsychological tests: Administration, norms, and commentary*. Oxford University Press.
- Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6), 643.
- Tamm, L., & Peugh, J. (2019). Concordance of teacher-rated and performance-based measures of executive functioning in preschoolers. *Child Neuropsychology*, 25(3), 410–424. <https://doi.org/10.1080/09297049.2018.1484085>
- Tan, A., Delgaty, L., Steward, K., & Bunner, M. (2018). Performance-based measures and behavioral ratings of executive function in diagnosing attention-deficit/hyperactivity disorder in children. *Attention Deficit and Hyperactivity Disorders*, 10, 309–316. <https://doi.org/10.1007/s12402-018-0256-y>
- Toplak, M. E., West, R. F., & Stanovich, K. E. (2013). Practitioner Review: Do performance-based measures and ratings of executive function assess the same construct? *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 54(2), 131–143. <https://doi.org/10.1111/jcpp.12001>
- Vaidya, C. J., You, X., Mostofsky, S., Pereira, F., Berl, M. M., & Kenworthy, L. (2020). Data-driven identification of subtypes of executive function across typical development, attention deficit hyperactivity disorder, and autism spectrum disorders. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, 61(1), 51–61. <https://doi.org/10.1111/jcpp.13114>
- Wechsler, D. (2005). *Wechsler Individual Achievement Test II (Second Edition)*. Pearson Assessment.
- Wehrens, R., & Buydens, L. M. C. (2007). Self- and super-organizing maps in R: The kohonen package. *Journal of Statistical Software*, 21, 1–19. <https://doi.org/10.18637/jss.v021.i05>
- Willcutt, E. G., Doyle, A. E., Nigg, J. T., Faraone, S. V., & Pennington, B. F. (2005). Validity of the executive function theory of attention-deficit/hyperactivity disorder: A meta-analytic review. *Biological Psychiatry*, 57(11), 1336–1346. <https://doi.org/10.1016/j.biopsych.2005.02.006>
- Williams, K. L., Holmes, J., Farina, F., Vedeckina, M., Bennett, M. P., & The CALM Team. (2022). Inconsistencies between subjective reports of cognitive difficulties and performance on cognitive tests are associated with elevated internalising and externalising symptoms in children with learning-related problems. *Research on Child and Adolescent Psychopathology*, 50(12), 1557–1572. <https://doi.org/10.1007/s10802-022-00930-4>
- Yeniad, N., Malda, M., Mesman, J., van IJendoorn, M. H., & Pieper, S. (2013). Shifting ability predicts math and reading performance in children: A meta-analytical study. *Learning and Individual Differences*, 23, 1–9. <https://doi.org/10.1016/j.lindif.2012.10.004>
- Yeo, B. T. T., Krienen, F. M., Sepulcre, J., Sabuncu, M. R., Lashkari, D., Hollinshead, M., Roffman, J. L., Smoller, J. W., Zöllei, L., Polimeni, J. R., Fisch, B., Liu, H., & Buckner, R. L. (2011). The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *Journal of Neurophysiology*, 106(3), 1125–1165. <https://doi.org/10.1152/jn.00338.2011>
- Zink, N., Lenartowicz, A., & Markett, S. (2021). A new era for executive function research: On the transition from centralized to distributed executive functioning. *Neuroscience and Biobehavioral Reviews*, 124, 235–244. <https://doi.org/10.1016/j.neubiorev.2021.02.011>